26900062 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEBTG302 - BIOPROCESS TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One Word Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. What is the importance of downstream processing in biotechnology? [U] / [CO4]
2. Give an example of an enrichment medium. [U] / [CO3]
3. What is the role of starch in amylase production? [K] / [CO4]
4. Name a common complex nitrogen source. [K] / [CO3]
5. What is aerobic fermentation? [K] / [CO1]
6. What is the next step after isolation of microorganism? [K] / [CO2]
7. State any two important uses of packed-bed bioreactors. [K] / [CO3]
8. Quote the main function of an inhibitor. [K] / [CO3]
9. Give an example for primary metabolite. [K] / [CO1]

10. Recite the name of method of sterilization that is suitable for heat-sensitive components. [K] / [CO3]

11. Which is the most common method used for obtaining pure culture? [K] / [CO2]

12. Identify the best source for amylase producers-(soil/water/air). [K] / [CO2]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

13. Describe the role of buffers in maintaining the pH of fermentation media. [K] / [CO3]

14. Explain the fed batch culture system. [U] / [CO1]

15. Explain the significance of hybridization in the production of new microbial metabolites. [U] / [CO2]

16. Explain the general procedure followed for the immobilisation of amylase. [U] / [CO4]

17. Define revertants and explain their importance in strain improvement with an example. [K] / [CO2]

18. Give disadvantages of SMF. [U] / [CO1]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

19. Discuss the principle and methodology of amylase purification using ammonium sulphate precipitation. [U] / [CO4]

20. Describe in detail the design, construction, and major components of a stirred-tank bioreactor. Explain the functions of each component. [K] / [CO3]

21. Define downstream processing and describe the various steps involved in it. [U] / [CO4]

22. Explain secondary screening with examples. [U] / [CO2]

[15x2 = 30]